LAB 2

AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

PUBLIC BUS LIVE TRACKING & CROWDING ESTIMATOR

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Section: A

PROBLEM STATEMENT #24 | Smart Cities, Transport & Logistics

Objective: Convert Lab 1 functional requirements into Agile Epics and User Stories, prioritize and estimate them, simulate two Scrum sprints, and analyse progress.

1. EPICS

EPIC 1 — LIVE BUS TRACKING

Enable commuters to track the real-time location of selected buses using GPS data.

EPIC 2 — BUS ETA & DELAY NOTIFICATION

Provide commuters with continuously updated bus arrival times and notify them when significant delays occur.

EPIC 3 — CROWDING ESTIMATION

Estimate and display the passenger crowding level of active buses using ticketing sensor data.

EPIC 4 — FLEET MONITORING & CONTROL

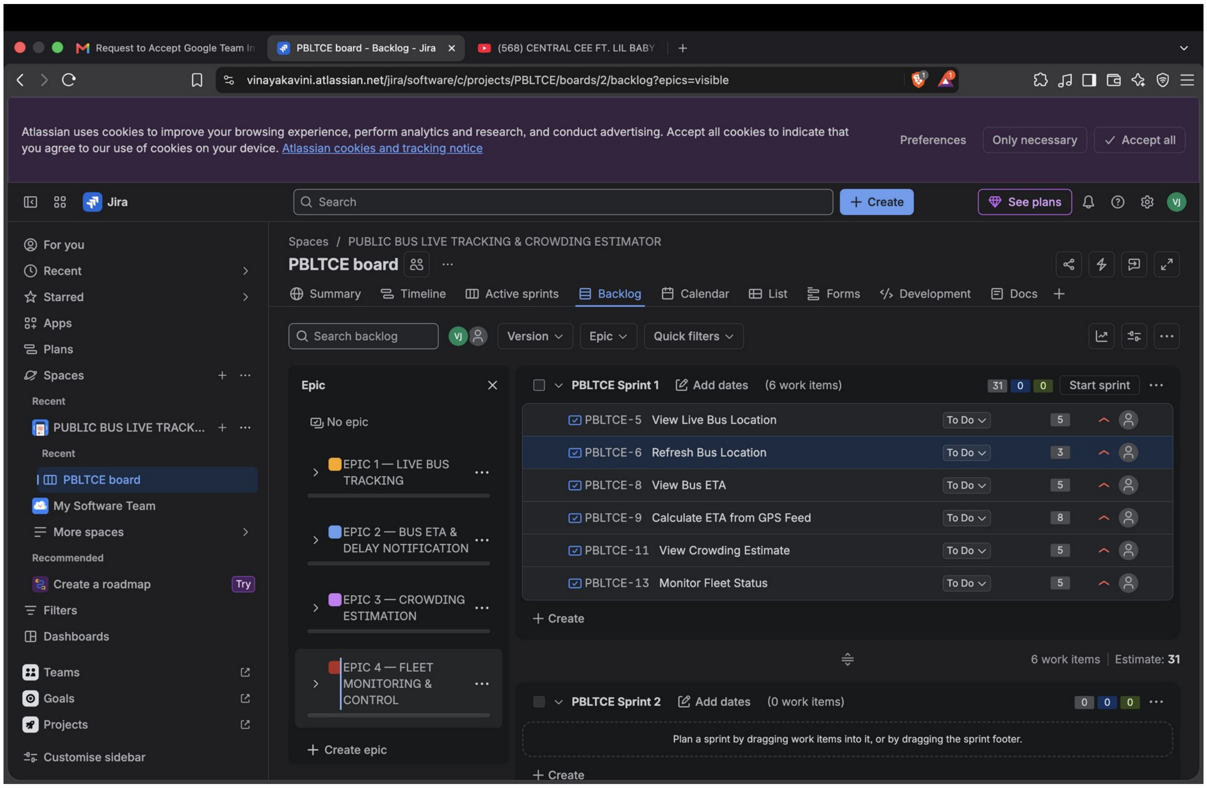
Provide fleet controllers with an operational dashboard and route-specific delay-alert configuration.

2. USER STORIES & STORY POINTS

EPIC	USER STORY	PRIORITY	POINTS
Live Bus Tracking	View Live Bus Location	High	5
Live Bus Tracking	Refresh Bus Location	High	3
Bus ETA & Delay	View Bus ETA	High	5
Bus ETA & Delay	Calculate ETA from GPS Feed	High	8
Bus ETA & Delay	Receive Delay Notification	Medium	5
Crowding Estimation	View Crowding Estimate	High	5
Crowding Estimation	Estimate Crowding Level from Sensors	High	8
Fleet Monitoring	Monitor Fleet Status	High	5
Fleet Monitoring	View Vehicle Route and Crowding	Medium	5
Fleet Monitoring	Configure Route Alert Thresholds	Medium	5

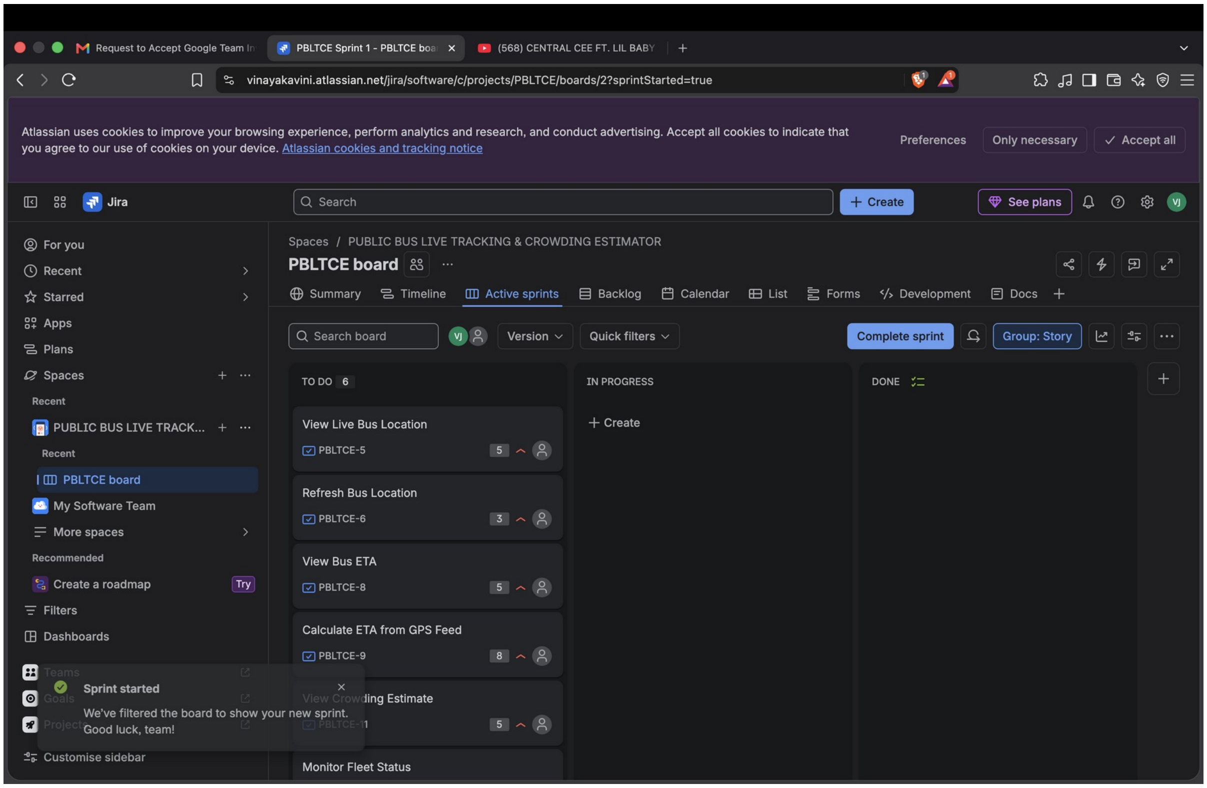
Total backlog estimate: 54 Story Points.

SCREENSHOT 1 — Jira Backlog with Epics and User Stories



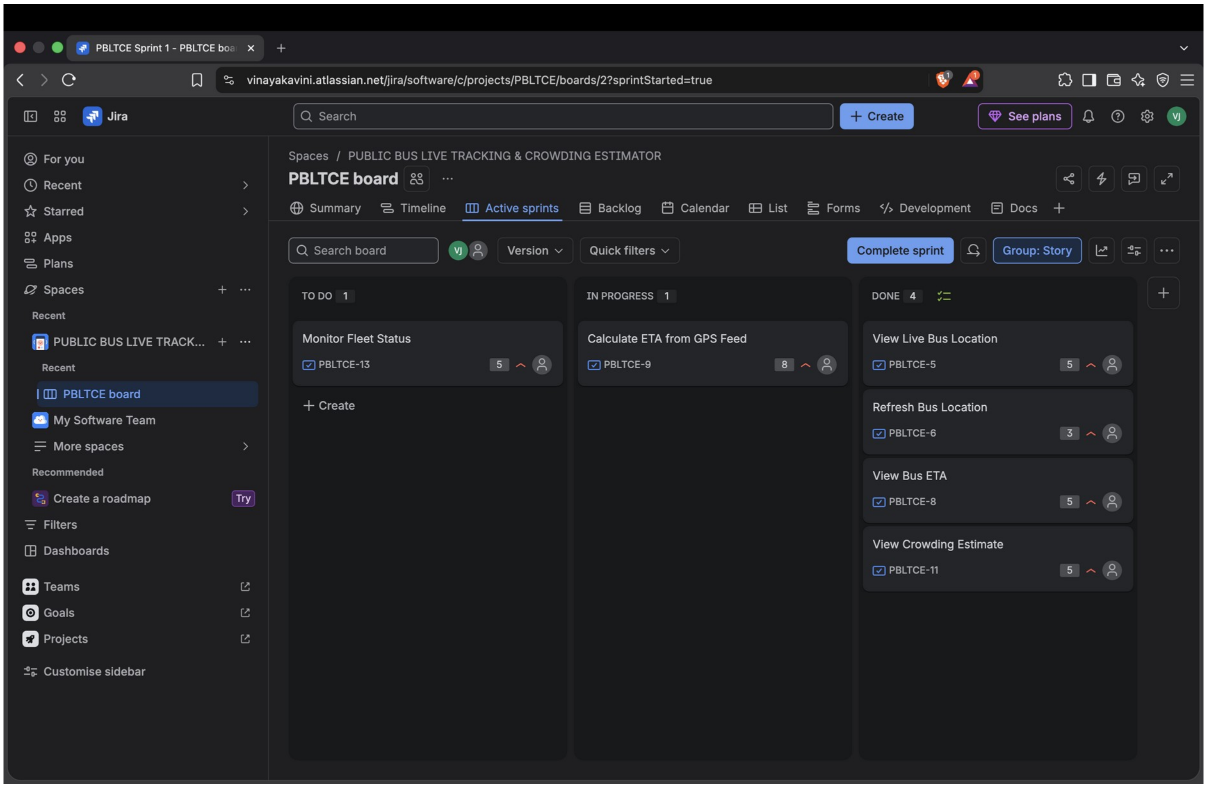
Screenshot 1 — Jira backlog with Epics and User Stories

SCREENSHOT 2 — Story Points & Priorities



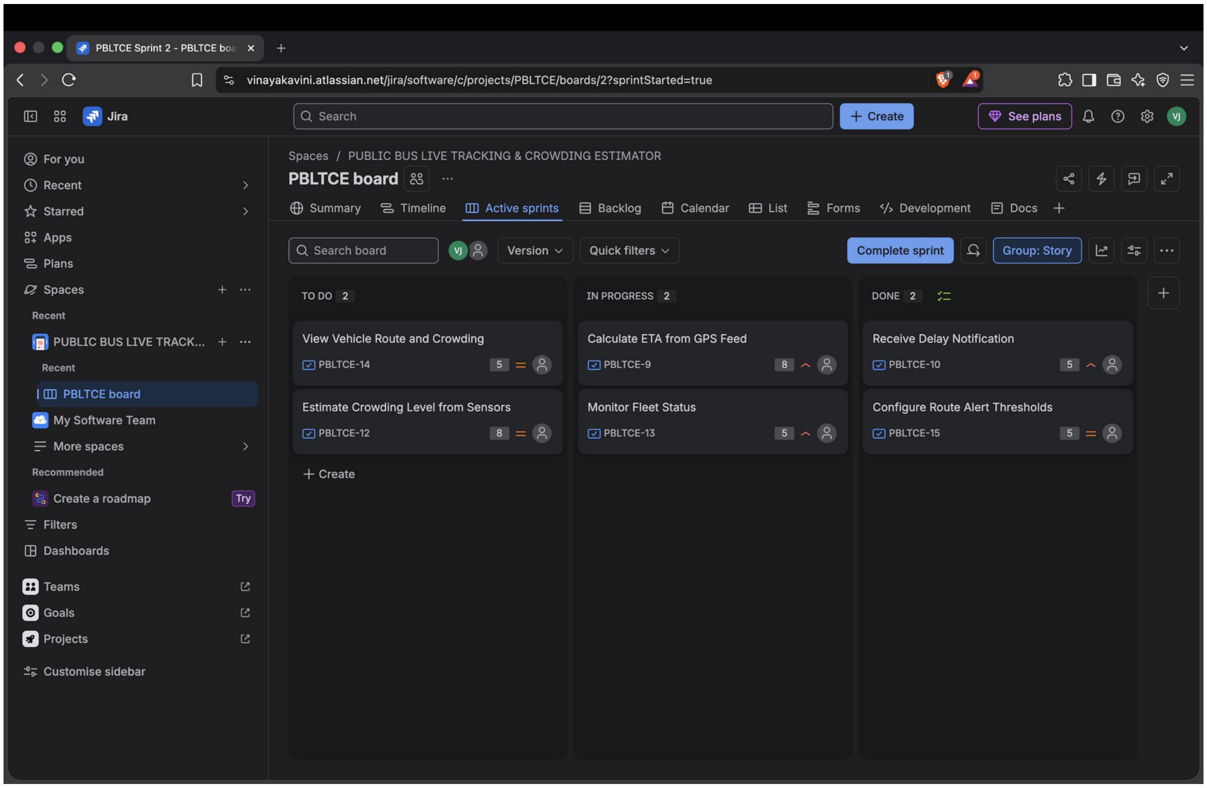
Screenshot 2 — Sprint 1 planned with six work items (31 Story Points)

SCREENSHOT 3 — Sprint 1: Active Sprint Board



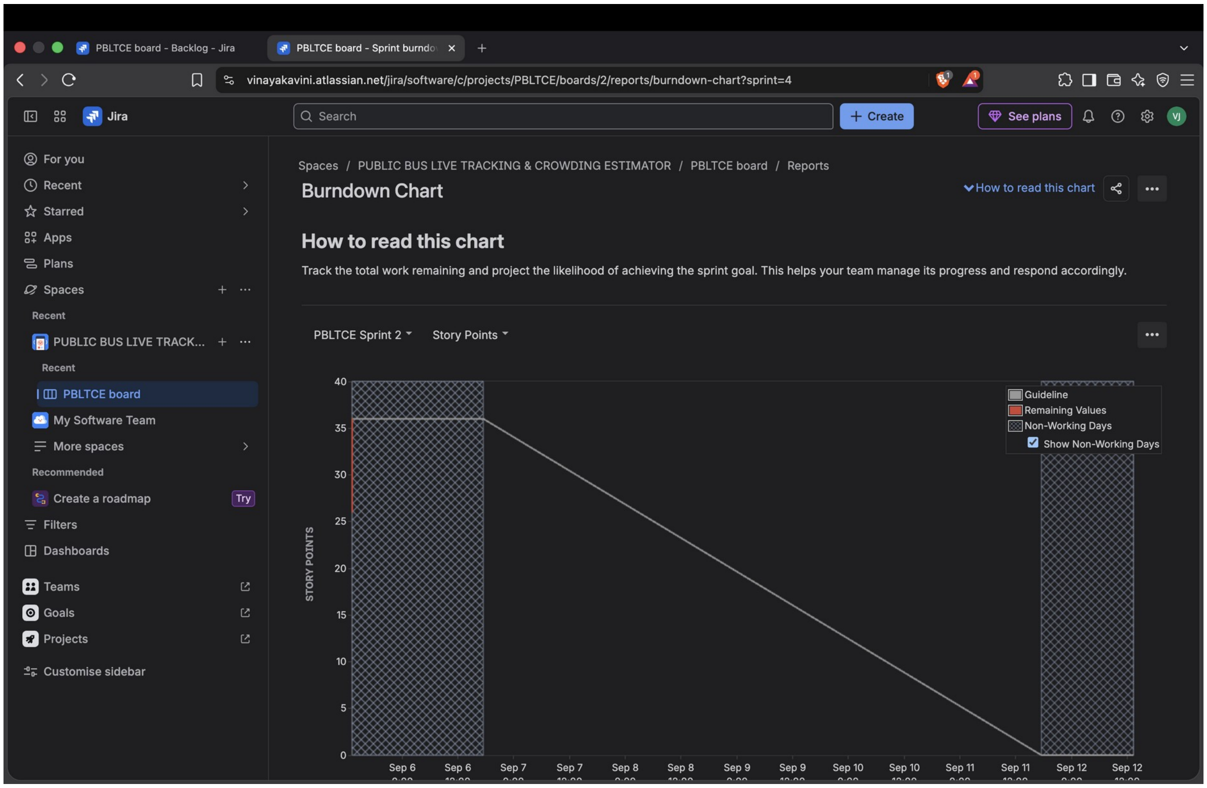
Screenshot 3 — Sprint 1 Active Sprint board: To Do / In Progress / Done

SCREENSHOT 4 — Sprint 2: Active Sprint Board



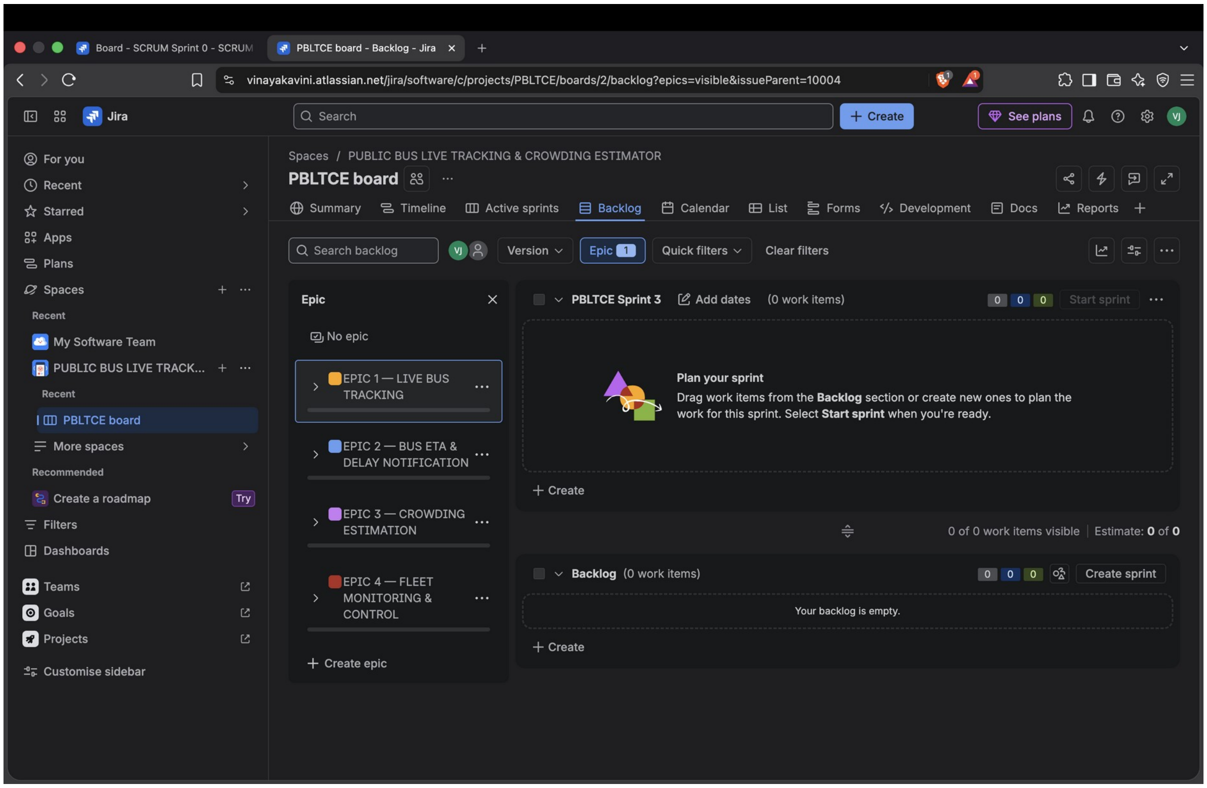
Screenshot 4 — Sprint 2 Active Sprint board

SCREENSHOT 5 — Burndown Chart



Screenshot 5 — Burndown Chart

SCREENSHOT 6 — Completed Sprint / Epic Progress



Screenshot 6 — Final Jira Backlog / Epic progress view

3. SPRINT SIMULATION SUMMARY

Sprint 1: View Live Bus Location (5), Refresh Bus Location (3), View Bus ETA (5), Calculate ETA from GPS Feed (8), View Crowding Estimate (5), and Monitor Fleet Status (5). Total = 31 Story Points.

Sprint 2: Receive Delay Notification (5), Estimate Crowding Level from Sensors (8), View Vehicle Route and Crowding (5), and Configure Route Alert Thresholds (5). Total = 23 Story Points.

The Lab 2 handout specifies a one-week sprint and requires finishing two sprints before generating/observing the Burndown Chart.

4. REFLECTION QUESTIONS

1. Did your estimations reflect the actual effort?

Yes. The story point estimates generally reflected the relative effort required for each task. Simpler tasks received fewer points, while ETA calculation and crowding estimation received higher points because they involved greater complexity and uncertainty.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized according to user impact and business value. Core commuter features such as live bus location, ETA and crowding information were given higher priority because they provide the main functionality of the system.

3. How did your simulated sprint align with your plan?

The simulated sprint followed the planned Agile workflow. The selected user stories were moved from To Do to In Progress and then to Done, allowing development progress to be simulated and evaluated.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed how quickly the team completed the planned work and how much work remained during the sprint. It helped illustrate the team's capacity and whether the planned story points could be completed within the sprint period.